

APPENDIX IV

Conversion Factors for Units

Charge:	1 coulomb = 2.998×10^9 statcoulomb
Current:	1 ampere $\equiv$ 1 coul/sec = 2.998×10^9 statampere
Potential:	1 volt = $(2.998 \times 10^2)^{-1}$ statvolt
Electric field:	1 volt/m = $(2.998 \times 10^4)^{-1}$ statvolt/cm
Magnetic induction:	1 weber/m ² $\equiv$ 1 tesla = 10^4 gauss
Resistance:	1 ohm = $(2.998)^{-2} \times 10^{-11}$ sec/cm
Conductivity:	1 mho/m = $(2.998)^2 \times 10^9$ sec ⁻¹
Capacitance:	1 farad = $(2.998)^2 \times 10^{11}$ cm
Magnetic flux:	1 weber = 10^8 gauss . cm ² (or maxwells)
Magnetic field:	1 ampere-turn/m = $4\pi \times 10^{-3}$ oersted
Force:	1 newton = 10^5 dyne
Energy:	1 joule = 10^7 erg
	1 electron volt (ev) = 1.602×10^{-19} joule
	1 ev = kT, where k is Boltzmann's constant, for T = 1.160×10^4 K
	1 Rydberg = 13.61 ev
Power:	1 watt $\equiv$ 1 joule/sec = 10^7 erg/sec
Pressure:	1 newton/m ² = 10 dyne/cm ²
	1 atm = 760 mm Hg = 1.013×10^5 newton/m ²
	1 torr = 1 mm Hg